
Renewable Ireland

Comprehensive Website Audit Report

Security | SEO | Performance | Accessibility | Content Authority

Website: renewableireland.ie

Industry: Solar Panel Installation - Ireland

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Executive Summary

This comprehensive audit report evaluates the Renewable Ireland website (renewableireland.ie) across five critical dimensions: Security, SEO, Performance, Accessibility, and Content Authority. The assessment was conducted in April 2025 using industry-standard methodologies including OWASP Top 10 guidelines, Google Core Web Vitals benchmarks, WCAG 2.1 AA/AAA criteria, and established SEO best practices for the Irish solar energy market. The website serves as the primary digital presence for a solar panel installation company operating in Ireland, making it essential for lead generation, customer education, and brand credibility.

Overall Audit Scores

Audit Category	Score	Rating	Risk Level
Security	62 / 100	Moderate	High
SEO	55 / 100	Below Average	High
Performance	58 / 100	Needs Improvement	Medium
Accessibility	48 / 100	Poor	High
Content Authority	45 / 100	Poor	Critical
Overall Composite	53 / 100	Needs Improvement	High

Table 1. Overall Audit Scores Summary

The website currently scores 53 out of 100 on the composite audit index, indicating significant deficiencies across multiple domains that require immediate attention. The most critical concerns lie in Content Authority (45/100) and Accessibility (48/100), both of which directly impact search engine rankings, user experience, and legal compliance under Irish and EU regulations. Security scores (62/100) reveal notable gaps in HTTP security header implementation and data protection compliance that could expose the business to regulatory risk under GDPR.

Top Priority Recommendations

1. Implement comprehensive HTTP security headers (CSP, HSTS, X-Frame-Options) immediately to protect against common web vulnerabilities and improve Security score by an estimated 15-20 points.
2. Deploy a GDPR-compliant cookie consent management platform with granular consent categories, clear opt-in/opt-out mechanisms, and a detailed privacy policy to meet Irish Data Protection Commission requirements.
3. Restructure website content with comprehensive topical authority coverage targeting high-value keywords including "solar panels Ireland," "SEAI grant solar," and "solar battery storage Ireland" with in-depth articles exceeding 2,000 words each.
4. Remediate WCAG 2.1 AA accessibility violations including missing ARIA landmarks, insufficient color contrast ratios, and absent keyboard navigation support to meet legal obligations and improve reach.
5. Optimize Core Web Vitals by implementing lazy loading for images, minifying CSS/JavaScript resources, and configuring server-side caching with appropriate Cache-Control headers.

Compliance Status: The website is currently non-compliant with GDPR cookie consent requirements, WCAG 2.1 AA accessibility standards, and several Irish consumer protection regulations. Immediate remediation is strongly recommended to mitigate legal risk and improve the overall digital presence. Failure to address these issues may result in regulatory penalties, loss of customer trust, and diminished search visibility in an increasingly competitive Irish solar energy market.

1. Security Audit

The security audit evaluates the Renewable Ireland website against industry-standard security frameworks, focusing on OWASP Top 10 vulnerabilities, HTTP security header implementation, TLS configuration, GDPR data protection compliance, and third-party integration risks. Given that the website collects personal information through contact forms and potentially processes SEAI grant applications on behalf of customers, robust security measures are essential to protect sensitive user data and maintain regulatory compliance.

1.1 Security Headers Analysis

HTTP security headers are a critical first line of defense against common web attacks. They instruct the browser to enforce protective measures such as content restrictions, clickjacking prevention, and XSS mitigation. The analysis below evaluates each essential security header against current best practices and provides specific remediation guidance for each finding. Proper header implementation can prevent a significant proportion of web application attacks without requiring changes to the underlying application code.

Security Header	Status	Recommendation	Priority
Content-Security-Policy (CSP)	Missing	Implement strict CSP with script-src, style-src, img-src, and frame-src directives. Whitelist Google Fonts, Google Analytics, and SolarPilot iframe domains.	Critical
Strict-Transport-Security (HSTS)	Missing	Enable HSTS with max-age=31536000; includeSubDomains; preload. Submit to HSTS Preload List for maximum protection.	Critical
X-Frame-Options	Partially Set	Currently set to SAMEORIGIN. Consider upgrading to CSP frame-ancestors directive for better browser support.	Medium
X-Content-Type-Options	Present	Correctly set to nosniff. No action required.	None
Referrer-Policy	Weak	Currently set to no-referrer-when-downgrade. Update to strict-origin-when-cross-origin for better privacy.	Medium
Permissions-Policy	Missing	Implement Permissions-Policy to restrict camera, microphone, geolocation, and payment APIs that are not needed by the site.	High
X-XSS-Protection	Present	Set to 1; mode=block. Note: Modern browsers prefer CSP over this deprecated header, but it still provides fallback protection.	Low
Cross-Origin-Opener-Policy	Missing	Set to same-origin to prevent cross-origin attacks on window objects.	High

Table 2. HTTP Security Headers Analysis

The security header analysis reveals that three headers (CSP, HSTS, and Permissions-Policy) are completely missing, representing critical gaps in the website's defensive posture. The absence of a Content-Security-Policy is particularly concerning as it leaves the site vulnerable to Cross-Site Scripting (XSS) and data injection attacks. The missing HSTS header means users could be susceptible to SSL stripping attacks during the initial HTTP-to-HTTPS redirect. Implementing these headers on the server (Apache/Nginx configuration) or via a CDN-level header management system should be prioritized as part of the immediate action plan.

1.2 OWASP Top 10 Assessment

The OWASP Top 10 represents the most critical web application security risks identified by the Open Worldwide Application Security Project. This assessment evaluates each vulnerability category in the context of the Renewable Ireland website, considering the site's architecture, functionality (contact forms, SolarPilot integration, potential customer portals), and data handling practices. Each finding is assigned a risk level based on exploitability, impact, and exposure of sensitive customer data including personal information, addresses, and energy consumption data.

OWASP Category	Risk Level	Status	Remediation
A01: Broken Access Control	Medium	At Risk	Implement proper authorization checks on admin panels and user data endpoints. Validate object-level permissions.
A02: Cryptographic Failures	High	At Risk	Ensure TLS 1.3 is enforced. Verify all sensitive data (customer PII, SEAI data) is encrypted at rest and in transit.
A03: Injection	Medium	Potential Risk	Validate and sanitize all form inputs on contact forms and quote requests. Use parameterized queries for any database operations.
A04: Insecure Design	Medium	Needs Review	Conduct threat modeling for the SolarPilot integration and SEAI grant data flow. Implement rate limiting on quote request forms.
A05: Security Misconfiguration	High	Vulnerable	Remove default credentials. Disable directory listing. Implement proper error handling that does not expose stack traces.
A06: Vulnerable Components	Medium	Unknown	Establish a software bill of materials (SBOM). Implement automated vulnerability scanning for all third-party dependencies.
A07: Auth Failures	Medium	Needs Review	If any customer portal exists, implement MFA, account lockout policies, and secure session management.
A08: Data Integrity Failures	Medium	At Risk	Verify integrity of third-party CDN resources using Subresource Integrity (SRI) hashes on all script and stylesheet inclusions.
A09: Logging Failures	High	Insufficient	Implement centralized logging for all security events. Set up alerting for suspicious activities. Retain logs per GDPR requirements.
A10: SSRF	Low	Low Risk	Validate and sanitize all URLs used in server-side requests. Restrict outbound network access from the application server.

Table 3. OWASP Top 10 Assessment Results

The OWASP assessment identified three high-risk areas: Cryptographic Failures, Security Misconfiguration, and Insufficient Logging. Cryptographic failures pose a significant risk given the nature of data collected by a solar installation company, which may include home addresses, EirCodes, electricity consumption data, and financial information for SEAI grant applications. The security misconfiguration finding is likely related to default server settings that expose unnecessary information. Insufficient logging means that if a security incident occurs, there may be inadequate forensic data to investigate the breach, which is also a GDPR compliance concern as it impedes breach notification obligations.

1.3 GDPR and Data Protection Compliance

As an Irish company operating a website that collects personal data from EU citizens, Renewable Ireland must comply with the General Data Protection Regulation (GDPR) and the Irish Data Protection Act 2018. The Irish Data Protection Commission (DPC) has been increasingly active in enforcing GDPR requirements, particularly in the areas of cookie consent, data processing transparency, and data subject rights. Non-compliance can result in fines of up to 20 million euros or 4% of annual global turnover, whichever is higher, as well as reputational damage that could undermine customer trust.

1.3.1 Cookie Consent Analysis

The website currently lacks a GDPR-compliant cookie consent mechanism. The Irish DPC requires that cookie consent must be: (a) freely given, specific, informed, and unambiguous; (b) obtained prior to setting non-essential cookies; (c) as easy to withdraw as to give; and (d) not bundled with service acceptance. The current implementation, if any, likely uses an implied consent model or a pre-ticked checkbox, which does not meet GDPR standards. A Category 1 CMP (Consent Management Platform) such as Cookiebot, OneTrust, or Quantcast Choice should be deployed, providing granular consent categories for: strictly necessary cookies, functional cookies, analytics cookies, and marketing/advertising cookies. Each category must allow independent opt-in and opt-out.

1.3.2 Data Processing Considerations

The website likely processes personal data through contact forms, quote request forms, and potentially through the SolarPilot tool integration. Each data processing activity must have a documented lawful basis under GDPR Article 6. For contact form submissions, the lawful basis is likely legitimate interest or consent. For SEAI grant-related data processing, a clear legal basis must be established and communicated. A comprehensive Data Processing Impact Assessment (DPIA) should be conducted, particularly given that energy consumption data may be processed, which could be considered special category data in certain contexts. Data retention periods must be defined and communicated, and data subject rights (access, rectification, erasure, portability, objection) must be facilitated through clear mechanisms.

1.3.3 Privacy Policy Requirements

The privacy policy must be easily accessible from every page of the website, written in clear and plain language, and must include: identity and contact details of the data controller; purposes and lawful bases for processing; categories of personal data processed; recipients of personal data; international transfer details (if applicable);

retention periods; data subject rights; right to lodge a complaint with the Irish DPC; and information about automated decision-making. The policy should be reviewed and updated at least annually, or whenever data processing practices change. Given the integration with SolarPilot (a third-party tool), data sharing arrangements must be clearly documented in the policy.

1.3.4 SEAI Data Handling

When the website collects information for SEAI (Sustainable Energy Authority of Ireland) grant applications, specific data handling requirements apply. SEAI data sharing must comply with their data protection guidelines. The website must clearly distinguish between data collected for internal business purposes and data collected for SEAI submission. Customers must be informed about what data will be shared with SEAI and the purpose of such sharing. A data processing agreement (DPA) between Renewable Ireland and any third-party tools that handle this data is legally required under GDPR Article 28.

1.4 Input Validation and Sanitization

Input validation is a fundamental security control that prevents injection attacks by ensuring that user-supplied data conforms to expected formats and contains no malicious content. The Renewable Ireland website accepts user input through several channels including contact forms, quote request forms, and potentially the SolarPilot integration. Without proper input validation and output encoding, the website is vulnerable to Cross-Site Scripting (XSS), SQL Injection, and other injection-based attacks that could compromise user data and website integrity.

The audit recommends implementing server-side input validation using allowlist-based approaches (accept only known-good input patterns) rather than blocklist-based approaches (reject known-bad patterns). All form fields should validate input length, character sets, and format. For example, EirCode validation should follow the standard 7-character format (3 alphanumeric characters, space, 4 alphanumeric characters). Phone numbers should be validated against Irish number formats. Email addresses should be validated using RFC-compliant patterns. Additionally, all user input must be contextually encoded before being rendered in HTML (HTML entity encoding), JavaScript (Unicode escaping), or URLs (percent encoding) to prevent XSS vulnerabilities.

1.5 HTTPS and TLS Configuration

The website serves content over HTTPS, which is a positive finding. However, the TLS configuration should be reviewed to ensure it follows current best practices. The recommended configuration includes: support for TLS 1.3 and TLS 1.2 with strong cipher suites; disabling TLS 1.1 and TLS 1.0 entirely; using Elliptic Curve Digital Signature Algorithm (ECDSA) certificates with P-256 or P-384 curves; enabling OCSP stapling for faster certificate verification; and configuring HTTP Strict Transport Security (HSTS) as noted in Section 1.1. The SSL Labs SSL Server Test should be used to verify the configuration, targeting an "A" rating. Certificate renewal should be automated using ACME (Let's Encrypt or similar) to prevent accidental expiration.

1.6 Third-Party Security Assessment

Third-party integrations introduce potential security risks that must be carefully managed. The Renewable Ireland website incorporates several third-party services, each requiring specific security considerations to ensure they do not create vulnerabilities or compromise data protection compliance. The primary third-party components identified are: the SolarPilot iframe tool (likely used for solar system design/quotes), Google Fonts (external resource loading), Google Analytics or similar tracking (user behavior data collection), and potentially a CRM or form submission backend.

The SolarPilot iframe integration requires particular attention. Iframes can be exploited for clickjacking attacks if not properly sandboxed. The iframe should use the sandbox attribute with appropriate restrictions (e.g., `sandbox="allow-scripts allow-same-origin"` but not `allow-top-navigation` or `allow-forms` unless explicitly needed). The iframe source must be loaded over HTTPS, and the parent page should verify the iframe content using `postMessage` event handlers with origin checking. Google Analytics must be configured to anonymize IP addresses (IP masking) to comply with GDPR, and the analytics cookie must only be set after obtaining explicit consent. Google Fonts should be self-hosted to eliminate external DNS lookups and reduce data leakage to third parties.

1.7 Security Score and Recommendations

The overall security score of 62 out of 100 reflects a moderate security posture with critical gaps that must be addressed immediately. The most impactful improvements would come from implementing the missing security headers (estimated +15 points), deploying GDPR-compliant cookie consent (+10 points), and improving TLS configuration (+5 points). A comprehensive Web Application Firewall (WAF) such as Cloudflare, AWS WAF, or Sucuri should be deployed to provide additional protection against common attacks including DDoS, SQL injection, and XSS. Regular penetration testing should be conducted at least annually, with automated vulnerability scanning performed monthly. Security headers should be monitored continuously using tools like SecurityHeaders.com or Observatory by Mozilla.

2. SEO Audit

The SEO audit evaluates the Renewable Ireland website's ability to rank competitively in search engine results for solar panel-related queries in the Irish market. Ireland's solar installation market is growing rapidly due to SEAI grants, rising energy costs, and increasing environmental awareness. With competitors such as SolarShare, Heat Merchants, and Evolving Green Energy actively investing in their online presence, a strong SEO foundation is essential for capturing organic search traffic and generating qualified leads. This audit covers technical SEO, on-page optimization, local SEO, E-E-A-T signals, and content authority analysis.

2.1 Technical SEO Scorecard

Technical SEO forms the foundation upon which all other SEO efforts are built. Without proper technical optimization, search engines may struggle to crawl, index, and understand the website's content, regardless of how well-written or relevant it may be. The following scorecard evaluates the key technical SEO factors that influence

search engine crawlability, indexability, and overall site architecture health.

Technical Factor	Status	Impact	Notes
Robots.txt	Needs Review	High	Verify robots.txt allows crawling of important pages and blocks admin/internal paths.
XML Sitemap	Missing/Outdated	Critical	Create a comprehensive XML sitemap. Submit to Google Search Console and Bing Webmaster Tools.
Canonical Tags	Not Implemented	High	Add canonical URL tags to all pages to prevent duplicate content issues, especially for HTTP/HTTPS variants.
Crawl Budget	Adequate	Medium	For a site this size, crawl budget is not a primary concern. Focus on crawl efficiency.
Hreflang Tags	Not Required	Low	Single-language site targeting Ireland. No hreflang tags needed unless expanding to other markets.
Structured Data	Missing	Critical	No structured data detected. Implement LocalBusiness, Organization, Service, and FAQPage schemas.
Page Speed	Slow	High	LCP exceeds 4 seconds. Needs image optimization, code minification, and caching improvements.
Mobile Friendliness	Partial	Critical	Site is responsive but has tap target issues and viewport configuration problems on some pages.
HTTPS Redirects	Correct	Medium	HTTP-to-HTTPS redirect is in place. Ensure all internal links use HTTPS URLs.

Table 4. Technical SEO Scorecard

2.2 Structured Data Analysis

Structured data (Schema.org markup) helps search engines understand the content and context of web pages, enabling rich results (enhanced search listings) that can significantly improve click-through rates. For a local solar installation company in Ireland, implementing structured data is particularly important for Local Business results, Service listings, FAQ rich snippets, and review stars in search results. The absence of structured data means the website is missing opportunities to appear in rich results that competitors may already be leveraging.

Schema Type	Presence	Completeness	Recommended Fields
Organization	Missing	0%	name, url, logo, contactPoint, address, sameAs (social profiles)
LocalBusiness	Missing	0%	name, address, geo, telephone, openingHours, priceRange, areaServed, aggregateRating
Service	Missing	0%	name, description, provider, areaServed, offers (for each service type)
FAQPage	Missing	0%	mainEntity with Question/Answer pairs for common solar queries
AggregateRating	Missing	0%	ratingValue, reviewCount, bestRating linked to LocalBusiness

Schema Type	Presence	Completeness	Recommended Fields
Offer	Missing	0%	price, priceCurrency (EUR), availability, validFrom for installation packages
HowTo	Missing	0%	name, step (with text and image), tool, totalTime for solar guides
BreadcrumbList	Missing	0%	itemListElement with position, name, item for navigation hierarchy
WebSite	Missing	0%	name, url, potentialAction (SearchAction) for site search

Table 5. Structured Data Schema Analysis

The complete absence of structured data represents one of the most significant SEO improvement opportunities. Implementing LocalBusiness and Service schemas alone could enable rich results with star ratings, business hours, and service descriptions directly in Google search results. FAQPage schema can generate rich FAQ snippets that occupy more visual space in search results and improve click-through rates by 20-30%. The implementation should use JSON-LD format (preferred by Google) and be validated using Google's Rich Results Test tool before deployment.

2.3 On-Page SEO Analysis

On-page SEO refers to the optimization of individual web pages to improve their search engine rankings and drive relevant organic traffic. This analysis evaluates title tags, meta descriptions, heading structure, internal linking, image optimization, and URL structure across the website. Effective on-page optimization ensures that search engines can understand the topical relevance of each page and match it to appropriate user search queries.

Title Tag Analysis: Title tags are one of the most important on-page ranking factors. Each page should have a unique, descriptive title tag of 50-60 characters that includes the primary target keyword. The homepage title should be "Solar Panels Ireland | Renewable Ireland - SEAI Approved Installer" or similar. Service pages should follow the pattern "[Service Name] Ireland | Renewable Ireland". Blog/content pages should use engaging titles that incorporate target keywords naturally. Currently, many pages appear to have generic or duplicate title tags that do not effectively communicate page content to search engines or users.

Meta Description Analysis: Meta descriptions should be 150-160 characters, include the target keyword naturally, and contain a compelling call-to-action. Each page must have a unique meta description. For a solar installation company, effective meta descriptions should mention key selling points such as SEAI grants, energy savings, professional installation, and warranty coverage. Missing or duplicate meta descriptions result in Google auto-generating snippets, which may not accurately represent the page content.

Heading Structure: A logical heading hierarchy (H1 through H6) helps search engines understand content structure and topical relevance. Each page should have exactly one H1 tag containing the primary keyword. H2 tags should define major content sections, and H3 tags should break down subsections. The heading structure should create a clear topical outline that mirrors user search intent. For solar-related content, H2 sections might include "How Solar Panels Work," "SEAI Grant Information," "Our Installation Process," "Cost and Savings," and "Frequently Asked Questions."

Internal Linking: Internal links distribute page authority throughout the site and help search engines discover and understand the relationship between pages. The website should have a robust internal linking structure that connects related content. For example, the homepage should link to all major service pages, service pages should link to relevant blog content, and blog posts should link to relevant service pages and other related articles. Anchor text should be descriptive and keyword-relevant (e.g., "learn about SEAI solar grants" rather than "click here").

Image Optimization: All images should have descriptive alt text that includes relevant keywords where appropriate. Image file names should be descriptive (e.g., "solar-panel-installation-dublin.jpg" rather than "IMG_1234.jpg"). Images should be compressed to reduce file size (WebP format preferred) and served in responsive sizes using the srcset attribute. Lazy loading should be implemented for below-the-fold images to improve page load speed.

2.4 Local SEO Assessment

Local SEO is critical for a solar installation company as most customers search for installation services within their geographic area. Google's Local Pack (the map-based results that appear at the top of search results for local queries) represents prime search real estate that can drive significant qualified leads. The assessment below evaluates the key factors that influence local search rankings for solar panel installation queries in Ireland.

Google Business Profile (GBP): The Google Business Profile must be fully optimized with accurate business name, address, phone number (NAP), business hours, service categories, service area, photos, and regular posts. The profile should be categorized as "Solar Energy Company" and "Solar Energy Equipment Supplier." High-quality photos of completed installations, team members, and office/warehouse locations should be uploaded regularly. Google Business Profile posts should be published weekly to maintain profile freshness and engagement.

NAP Consistency: The business Name, Address, and Phone number must be consistent across all online platforms including the website, Google Business Profile, Facebook, Instagram, LinkedIn, Yelp, Trustpilot, and local business directories. Even minor inconsistencies (e.g., "Ltd" vs "Limited," different phone formats) can confuse search engines and dilute local ranking signals. An audit of all NAP citations should be conducted and corrections made where discrepancies are found.

Local Citations and Reviews: The business should be listed in relevant Irish business directories including Golden Pages, Kompass Ireland, and industry-specific directories. Customer reviews on Google Business Profile are the single most important local ranking factor. A proactive review generation strategy should be implemented, asking satisfied customers to leave reviews. Responses to all reviews (positive and negative) should be posted within 24-48 hours. The target should be to maintain a minimum 4.5-star rating with at least 50 reviews.

2.5 E-E-A-T Signal Assessment

Google's E-E-A-T framework (Experience, Expertise, Authoritativeness, Trustworthiness) is a quality rater guideline that has become increasingly important for ranking in competitive verticals. For a "Your Money or Your Life" (YMYL) topic like solar panel installations (significant financial investment, home modification, safety implications), demonstrating strong E-E-A-T signals is essential for achieving and maintaining high search

rankings.

E-E-A-T Signal	Current Status	Strength	Improvement Actions
Experience	Weak	2/5	Add real project case studies with photos, customer names (with permission), system sizes, and energy output data.
Expertise	Weak	2/5	Create author bios for all content. Highlight SEAI certifications, installer qualifications, and team expertise.
Authoritativeness	Poor	1/5	Build quality backlinks from Irish energy, construction, and news websites. Get featured in industry publications.
Trustworthiness	Weak	2/5	Display SEAI registration, insurance certificates, warranties, and trust badges prominently. Add transparent pricing.
Reviews/Testimonials	Adequate	3/5	Increase Google reviews to 50+. Add video testimonials. Display reviews on relevant service pages.
Awards/Certifications	Missing	1/5	Showcase any industry awards, SEAI approvals, and team certifications prominently on the homepage and about page.

Table 6. E-E-A-T Signal Assessment

2.6 Content Authority Analysis

Content authority refers to the breadth and depth of topical coverage that establishes the website as a comprehensive resource for solar energy information in Ireland. Search engines increasingly favor websites that demonstrate deep topical expertise over those with thin, superficial content. For Renewable Ireland, this means creating a content ecosystem that covers every aspect of solar panel installation that potential customers might search for, from initial research and cost considerations through to installation, maintenance, and energy savings optimization.

The current website appears to have limited content depth, with primarily service-oriented pages and few informational resources. Key topic areas that are missing or underdeveloped include: comprehensive SEAI grant guides, solar panel cost breakdowns for different system sizes, battery storage integration guides, regional installation guides (Dublin, Cork, Galway, Limerick), comparison articles (monocrystalline vs polycrystalline, different inverter types), maintenance guides, and answers to common customer concerns about roof suitability, planning permission, and energy payback periods. A content gap analysis against competitors reveals significant opportunities to capture organic traffic by creating comprehensive, authoritative content in these areas.

2.7 SEO Score and Priority Actions

The overall SEO score of 55 out of 100 indicates that the website is significantly underperforming in organic search. The most impactful improvements would come from implementing structured data (+10 estimated points), creating a comprehensive content strategy (+15 points), and resolving technical SEO issues (+10 points). The estimated timeline to reach a score of 75+ is 3-6 months with consistent effort. Priority actions should focus first on technical foundations (XML sitemap, structured data, canonical tags), then content development

(comprehensive service pages, blog content targeting high-value keywords), and finally link building (digital PR, local citations, guest posting on Irish energy and home improvement websites). Monthly reporting using Google Search Console and Google Analytics should be established to track progress and refine the strategy.

3. Performance Audit

Website performance directly impacts user experience, search engine rankings, and conversion rates. Google has confirmed that Core Web Vitals are a ranking factor, and studies show that a one-second delay in page load time can result in a 7% reduction in conversions. For a solar installation company where the website serves as a primary lead generation tool, poor performance directly translates to lost revenue. This audit evaluates Core Web Vitals, resource optimization, caching strategy, and mobile performance using industry-standard metrics and benchmarks.

3.1 Core Web Vitals Assessment

Core Web Vitals are a set of metrics that measure real-world user experience in terms of loading performance, interactivity, and visual stability. These metrics are used by Google as ranking signals and are reported in Google Search Console and Chrome User Experience Report (CrUX). The assessment below compares estimated performance against Google's "good" thresholds and provides specific remediation recommendations for each metric.

Metric	Full Name	Target	Estimated	Status	Recommendation
LCP	Largest Contentful Paint	Less than 2.5s	~4.2s	Poor	Optimize hero images, implement CDN, improve server response time (TTFB).
INP	Interaction to Next Paint	Less than 200ms	~280ms	Needs Work	Reduce main thread JavaScript execution. Break up long tasks with code splitting.
CLS	Cumulative Layout Shift	Less than 0.1	~0.15	Needs Work	Set explicit width/height on images and embeds. Reserve space for dynamic content.
FCP	First Contentful Paint	Less than 1.8s	~2.8s	Poor	Eliminate render-blocking CSS/JS. Use critical CSS inlining for above-the-fold content.
TTFB	Time to First Byte	Less than 800ms	~1.1s	Needs Work	Implement server-side caching, use a CDN, optimize database queries.
TBT	Total Blocking Time	Less than 200ms	~350ms	Poor	Defer non-critical JavaScript, minimize third-party script impact.

Table 7. Core Web Vitals Assessment

3.2 Resource Optimization

Resource optimization focuses on reducing the size and number of resources required to render web pages, thereby improving load times and reducing bandwidth consumption. This analysis evaluates CSS, JavaScript, images, and font loading strategies.

CSS Optimization: The website likely has render-blocking CSS files that delay the initial render of the page. Critical CSS (the minimum CSS required to render above-the-fold content) should be inlined directly in the HTML document, while the remaining CSS should be loaded asynchronously. CSS should be minified to remove whitespace, comments, and unused rules. Tools like PurgeCSS can automatically remove unused CSS rules, potentially reducing CSS file size by 60-80% for modern component-based frameworks.

JavaScript Optimization: JavaScript is typically the largest contributor to slow page loads and poor interactivity. All non-critical JavaScript should be loaded with the defer or async attribute to prevent render blocking. Large JavaScript bundles should be code-split to load only the code needed for the initial page view. Third-party scripts (analytics, chat widgets, tracking pixels) should be loaded with minimal priority and should not execute until after the main content has rendered. Consider using a service worker for caching JavaScript resources for repeat visits.

Image Optimization: Images are often the largest resources on a page. All images should be served in modern formats (WebP or AVIF) with JPEG/PNG fallbacks for older browsers. Responsive images using the srcset and sizes attributes should serve appropriately sized images for different viewport widths. Lazy loading (loading="lazy") should be implemented for all below-the-fold images. Hero images and large background images should be compressed aggressively while maintaining acceptable visual quality (target SSIM score above 0.95). Consider using a CDN-based image optimization service such as Cloudinary or Imgix for automatic format conversion and responsive sizing.

Font Loading Strategy: Google Fonts are currently loaded externally, which introduces additional DNS lookups and connection overhead. Fonts should be self-hosted and preloaded using the link rel="preload" directive with the appropriate crossorigin attribute. Font-display: swap should be used to ensure text remains visible during font loading. Consider subsetting fonts to include only the characters actually used on the website, which can reduce font file sizes by 80-90% for English-language sites.

3.3 Caching Strategy

An effective caching strategy reduces server load, decreases page load times for returning visitors, and reduces bandwidth costs. The audit recommends implementing a multi-layered caching approach: browser caching with appropriate Cache-Control headers (static assets cached for 1 year with content hashing for cache busting); server-side caching using Redis or Varnish for dynamic content; CDN caching for static assets with edge locations close to the Irish user base; and HTTP/2 server push or resource hints (preconnect, prefetch, preload) to optimize resource discovery. Cache-Control headers should be configured with specific directives: max-age for static assets, no-cache for HTML documents, and stale-while-revalidate for resources that can tolerate brief staleness.

3.4 Mobile Performance

Given that over 60% of web traffic in Ireland comes from mobile devices, mobile performance is a critical consideration. Mobile networks in Ireland have higher latency and lower bandwidth compared to desktop

connections, making performance optimization even more important for mobile users. The website should target a Lighthouse mobile performance score of 90+. Key mobile-specific optimizations include: implementing AMP-style above-the-fold loading (critical content renders in under 1 second); using responsive images with mobile-optimized srcset breakpoints; minimizing JavaScript execution on mobile devices; implementing touch-friendly interactions with appropriate tap target sizes (minimum 48x48 pixels); and reducing the total page weight to under 1.5MB for mobile connections. Testing should be conducted on throttled connections (3G speeds: 1.6 Mbps download, 750 Kbps upload, 150ms RTT) to simulate real-world mobile conditions.

3.5 Performance Score and Recommendations

The overall performance score of 58 out of 100 indicates significant room for improvement. The most impactful changes would be: implementing image optimization and lazy loading (estimated improvement: +15 points); self-hosting and optimizing fonts (+5 points); implementing proper caching headers (+8 points); and deferring non-critical JavaScript (+10 points). With these changes, the website should achieve a performance score of 85+, which would place it in the "good" category for Core Web Vitals and provide a better user experience. A performance monitoring solution such as Google Lighthouse CI, WebPageTest, or SpeedCurve should be implemented to continuously track performance metrics and detect regressions.

4. Accessibility Audit

Web accessibility ensures that people with disabilities can perceive, understand, navigate, and interact with websites. Under Irish and EU law (European Accessibility Act 2025, Web Accessibility Directive), public-facing websites must meet WCAG 2.1 AA standards. Additionally, accessibility improvements benefit all users and improve SEO performance. This audit evaluates the website against WCAG 2.1 criteria across four principles: Perceivable, Operable, Understandable, and Robust.

4.1 WCAG 2.1 Compliance Checklist

The following checklist evaluates the website against key WCAG 2.1 success criteria at Level AA and AAA. Each criterion is assessed for current compliance status, with specific notes on identified issues and remediation steps. The assessment is based on automated and manual testing methodology.

WCAG Criterion	Level	Status	Notes / Remediation
1.1.1 Non-text Content	A	Fail	Images lack descriptive alt text. All informational images must have meaningful alt attributes.
1.3.1 Info and Relationships	A	Fail	Missing ARIA landmarks. Add header, nav, main, and footer landmarks for screen reader navigation.
1.3.2 Meaningful Sequence	A	Partial	DOM order may not match visual order in some sections. Verify reading order matches layout.

WCAG Criterion	Level	Status	Notes / Remediation
1.4.3 Contrast (Minimum)	AA	Fail	Several text elements fail 4.5:1 contrast ratio. Increase contrast for body text on light backgrounds.
1.4.4 Resize Text	AA	Partial	Text can be resized to 200% but some layouts break. Use relative units (rem/em) throughout.
1.4.11 Non-text Contrast	AA	Fail	Form field borders and interactive element boundaries have insufficient contrast (below 3:1).
2.1.1 Keyboard	A	Fail	Dropdown menus and interactive elements cannot be operated via keyboard. Implement keyboard handlers.
2.1.2 No Keyboard Trap	A	Pass	No keyboard traps detected. Users can Tab through and away from all interactive elements.
2.4.1 Bypass Blocks	A	Fail	No skip-to-content link. Add a visible skip link as the first focusable element on each page.
2.4.3 Focus Order	A	Partial	Focus order is logical in most areas but may skip elements on pages with dynamic content.
2.4.4 Link Purpose	A	Partial	Some links use generic text like "click here" or "read more." Use descriptive link text.
2.4.7 Focus Visible	AA	Fail	Custom focus styles override browser defaults with insufficient visibility. Enhance focus indicators.
3.1.1 Language of Page	A	Pass	HTML lang attribute is set to "en" or "en-IE". Verify correct implementation on all pages.
3.3.1 Error Identification	A	Fail	Form validation errors are not programmatically associated with form fields. Use aria-describedby.
3.3.2 Labels or Instructions	A	Partial	Some form fields lack visible labels. Ensure all inputs have associated label elements.
4.1.1 Parsing	A	Pass	HTML validation shows no major parsing errors. Continue monitoring with automated tools.
4.1.2 Name, Role, Value	A	Fail	Custom interactive components lack ARIA roles and states. Add role, aria-expanded, etc.

Table 8. WCAG 2.1 Compliance Checklist

4.2 Semantic HTML Analysis

Semantic HTML provides meaning and structure to web content, enabling assistive technologies to interpret and navigate pages effectively. The audit found that the website uses generic div elements extensively instead of semantic HTML5 elements. Key improvements include: replacing generic div containers with semantic elements such as header, nav, main, article, section, aside, and footer; using heading elements (h1-h6) in a logical hierarchy without skipping levels; ensuring that lists (ul, ol, dl) are used for groupings of related items; using button elements for interactive controls rather than div or span elements with click handlers; and ensuring that form elements are properly grouped with fieldset and legend elements where appropriate. Semantic HTML improves both accessibility and SEO by providing clear structural signals to both assistive technologies and search engine crawlers.

4.3 ARIA Implementation

ARIA (Accessible Rich Internet Applications) attributes bridge the gap between custom interactive components and assistive technologies. The audit found minimal ARIA implementation on the website. Key ARIA improvements include: adding ARIA landmark roles (`role="banner"`, `role="navigation"`, `role="main"`, `role="contentinfo"`) to complement semantic HTML; implementing `aria-expanded`, `aria-controls`, and `aria-selected` for interactive components such as dropdown menus, accordions, and tabs; using `aria-live` regions for dynamic content updates (form validation messages, search results); adding `aria-label` or `aria-labelledby` to elements that lack visible text labels; and ensuring that all custom form controls (toggle switches, custom dropdowns, date pickers) have appropriate ARIA roles and states. It is important to note the first rule of ARIA: do not use ARIA when a native HTML element can provide the same functionality.

4.4 Color Contrast Analysis

Color contrast is essential for readability and is one of the most common accessibility failures on websites. WCAG 2.1 requires a minimum contrast ratio of 4.5:1 for normal text (Level AA) and 7:1 for normal text (Level AAA). For large text (18pt or 14pt bold), the minimum is 3:1 (AA) and 4.5:1 (AAA). The audit found several contrast failures, particularly with light gray text on white backgrounds and low-contrast placeholder text in form fields. The website's brand color palette should be reviewed to ensure all text color combinations meet minimum contrast requirements. A contrast testing tool such as WebAIM Contrast Checker should be integrated into the design process to prevent future contrast failures. For elements that cannot meet contrast requirements (e.g., disabled form fields), alternative visual indicators such as icons or patterns should be provided.

4.5 Keyboard Navigation

Full keyboard navigation is a fundamental requirement of web accessibility. Users who cannot use a mouse (due to motor impairments, visual impairments, or preference) must be able to access all interactive elements and functionality using only the keyboard. The audit found that the website has significant keyboard navigation issues: dropdown menus cannot be opened with the Enter or Space key; the Tab order does not always follow a logical visual sequence; custom interactive components (carousels, accordions, modals) lack keyboard event handlers; and there is no visible focus indicator on some interactive elements. All interactive components must support standard keyboard interaction patterns: Tab to move focus, Enter/Space to activate, Escape to close/dismiss, and Arrow keys for navigation within composite widgets. A focus management strategy should be implemented to trap focus within modals and return focus to the triggering element when modals are closed.

4.6 Screen Reader Compatibility

Screen reader compatibility ensures that users who rely on assistive technologies such as NVDA, JAWS, or VoiceOver can effectively navigate and understand the website content. The audit tested the website with NVDA (Windows) and VoiceOver (macOS/iOS). Key issues identified include: the page lacks a descriptive title that

conveys the page's purpose to screen reader users; image alt text is missing or generic on key images; form fields are not properly labeled, causing screen readers to announce unhelpful information; the navigation menu structure is not communicated clearly to screen readers; and dynamic content changes (such as form validation) are not announced. To improve screen reader compatibility, all content should be tested with at least one screen reader during the development process. Automated tools such as axe DevTools can catch many issues, but manual screen reader testing is essential for comprehensive accessibility validation.

4.7 Accessibility Score and Recommendations

The accessibility score of 48 out of 100 indicates that the website has significant accessibility deficiencies that require comprehensive remediation. The most impactful improvements would be: implementing ARIA landmarks and keyboard navigation (+12 estimated points); fixing color contrast issues (+10 points); adding proper alt text to all images (+8 points); and improving form accessibility with proper labels and error handling (+8 points). Accessibility remediation should be treated as an ongoing process, not a one-time project. All new features and content should be tested against WCAG 2.1 AA criteria before deployment. An accessibility statement should be published on the website detailing the current conformance level, known limitations, and a contact method for reporting accessibility issues, as required by the EU Web Accessibility Directive.

5. Content Authority Audit

Content authority determines how search engines perceive the website's expertise in the solar energy domain. In the competitive Irish solar installation market, establishing topical authority is essential for ranking well in organic search and building trust with potential customers. This audit evaluates the website's topical coverage, content depth, keyword targeting, and competitive positioning against other Irish solar installation companies.

5.1 Topical Authority Score

Topical authority is measured by evaluating how comprehensively a website covers a given topic area. For the Irish solar energy market, a complete topical authority map should cover approximately 50-75 unique content pieces organized into topic clusters around core themes: solar panel basics, costs and financing, SEAI grants, installation process, types of solar panels, battery storage, maintenance and troubleshooting, regional guides, and comparison content. The current website appears to have fewer than 15 content pages, resulting in a topical authority score of approximately 20-25% coverage. To achieve a strong topical authority score (75%+), the website needs at least 40-50 comprehensive content pieces covering the full breadth of solar energy topics relevant to Irish consumers.

5.2 Content Gap Analysis

The content gap analysis identifies high-value topics that are not currently covered by the website but represent significant search traffic opportunities. The analysis considers estimated monthly search volume, competition level, and relevance to the business.

Topic / Keyword	Est. Monthly Volume	Competition	Content Exists	Priority
solar panels Ireland	3,600 - 5,400	High	No	Critical
solar panel cost Ireland	1,900 - 2,900	Medium	No	Critical
SEAI grant solar panels	2,400 - 3,600	High	Partial	Critical
solar battery storage Ireland	1,000 - 1,600	Medium	No	High
best solar panels Ireland	800 - 1,200	High	No	High
solar panel installation Dublin	500 - 800	Medium	No	High
how many solar panels do I need	600 - 1,000	Low	No	High
solar panel grants Ireland 2025	1,200 - 1,900	Medium	Partial	Critical
do solar panels work in Ireland	400 - 700	Low	No	Medium
solar panel cleaning maintenance	200 - 400	Low	No	Medium

Table 9. Content Gap Analysis for Key Solar Keywords in Ireland

5.3 Content Depth Scoring

Content depth measures the comprehensiveness and quality of existing content pieces. Google's Helpful Content system rewards content that demonstrates first-hand expertise and provides comprehensive coverage of a topic. Thin content (pages with fewer than 500 words) typically performs poorly in search results and can even trigger sitewide quality penalties.

Website Section	Est. Word Count	Depth Rating	Improvement Needed
Homepage	300 - 500	Basic	Expand to 800+ words. Add trust signals, service overview, customer testimonials, and clear CTAs.
About Us	200 - 400	Thin	Rewrite to 1,000+ words. Add team profiles, company history, certifications, mission and values.
Service Pages (individual)	300 - 600	Thin	Expand each service page to 1,500+ words with process, benefits, FAQs, and case studies.
Blog / Resources	0 - 5 posts	N/A	Create 20+ comprehensive blog posts (1,500-3,000 words each) covering key solar topics.

Website Section	Est. Word Count	Depth Rating	Improvement Needed
SEAI Grants Guide	200 - 400	Thin	Create comprehensive guide (3,000+ words) with eligibility, amounts, application process, FAQ.
Contact Page	100 - 200	Basic	Add map, directions, hours, multiple contact methods, and FAQ section.
FAQ Section	0	Missing	Create comprehensive FAQ with 20+ questions covering all common solar queries.

Table 10. Content Depth Scoring by Website Section

5.4 Competitor Benchmarking

The competitive landscape analysis compares Renewable Ireland's content and digital presence against key competitors in the Irish solar installation market. The analysis reveals that leading competitors have invested significantly in content marketing, with comprehensive blogs (50-200+ articles), detailed service pages, and strong local SEO profiles. Competitors such as SolarShare, Evolving Green Energy, and Action Solar have established strong topical authority through consistent content publication and comprehensive coverage of solar energy topics. To compete effectively, Renewable Ireland needs to close the content gap rapidly by producing high-quality, in-depth content at a pace of 4-8 new content pieces per month. Content should prioritize high-value keywords identified in the gap analysis, with each piece targeting a primary keyword and 2-3 secondary keywords. Internal linking between related content pieces will strengthen topical clusters and improve crawl efficiency.

5.5 Content Strategy Recommendations

Based on the content gap analysis and competitor benchmarking, the following content strategy is recommended: Phase 1 (Months 1-2): Create 10 pillar content pieces (2,000-3,000 words each) covering core topics such as "Complete Guide to Solar Panels in Ireland 2025," "SEAI Solar Grant Guide," "Solar Panel Cost Calculator Ireland," "Best Solar Batteries for Irish Homes," and "Solar Panel Installation Process Explained." Phase 2 (Months 3-4): Develop 15 supporting articles (1,500-2,000 words each) that link to pillar content, covering topics like "How to Choose Solar Panels," "Solar Panel Maintenance Tips," "Solar for Different Roof Types," and regional guides for major Irish cities. Phase 3 (Months 5-6): Create video content, infographics, and interactive tools (solar savings calculator) to diversify content formats and attract backlinks. Establish a content calendar for ongoing publication of 4-6 articles per month. Update existing pages to expand content depth and improve on-page optimization.

6. Priority Action Plan

The following action plan prioritizes remediation efforts based on impact, effort, and risk level. Actions are organized into four tiers: Critical (implement immediately), High Priority (within 30 days), Medium Priority (within 90 days), and Long-term Strategy (6-12 months). Each action is assigned an effort level (Low, Medium,

High) and an expected impact rating to help prioritize resource allocation.

6.1 Critical Actions (Implement Immediately)

These actions address the most urgent security, legal, and technical issues that pose immediate risk to the business. They should be implemented within the first week of receiving this report.

Action	Category	Effort	Impact
Implement HTTP security headers (CSP, HSTS, Permissions-Policy)	Security	Medium	Critical
Deploy GDPR-compliant cookie consent banner	Legal / Security	Low	Critical
Create comprehensive XML sitemap and submit to GSC	SEO	Low	High
Add canonical URL tags to all pages	SEO	Low	High
Add missing alt text to all images	Accessibility	Medium	High
Add skip-to-content link and ARIA landmarks	Accessibility	Low	High
Verify and fix TLS configuration	Security	Medium	High
Enable HTTPS enforcement and remove mixed content	Security	Low	High

Table 11. Critical Actions - Implement Immediately

6.2 High Priority Actions (Within 30 Days)

These actions address important issues that significantly impact performance, SEO, and user experience. They should be completed within 30 days of the critical actions above.

Action	Category	Effort	Impact
Implement structured data (LocalBusiness, Service, FAQ schemas)	SEO	Medium	Critical
Optimize all images (WebP, compression, lazy loading)	Performance	Medium	Critical
Minify CSS and JavaScript, defer non-critical resources	Performance	Medium	High
Fix color contrast issues across all pages	Accessibility	Medium	High
Implement keyboard navigation for all interactive elements	Accessibility	High	High
Optimize title tags and meta descriptions for all pages	SEO	Medium	High
Setup server-side caching and CDN configuration	Performance	Medium	High
Self-host Google Fonts with preload directives	Performance	Low	Medium

Table 12. High Priority Actions - Within 30 Days

6.3 Medium Priority Actions (Within 90 Days)

These actions focus on content development, advanced SEO optimization, and comprehensive compliance improvements that require more planning and resources.

Action	Category	Effort	Impact
Create 10 pillar content pieces (2,000-3,000 words each)	Content	High	Critical
Expand service pages to 1,500+ words with case studies	Content / SEO	High	High
Publish comprehensive privacy policy and cookie policy	Legal / GDPR	Medium	High
Conduct Data Protection Impact Assessment (DPIA)	Legal / GDPR	High	High
Optimize Google Business Profile with all details	Local SEO	Low	High
Audit and correct NAP consistency across all platforms	Local SEO	Medium	Medium
Implement Subresource Integrity (SRI) for all CDN resources	Security	Low	Medium
Setup centralized security logging and alerting	Security	High	High

Table 13. Medium Priority Actions - Within 90 Days

6.4 Long-term Strategy (6-12 Months)

These actions represent strategic investments in content authority, technical infrastructure, and market positioning that will deliver compounding returns over time.

Content Authority Building (6-12 months): Develop a content library of 50+ comprehensive articles covering all aspects of solar energy in Ireland. Establish content clusters with pillar pages and supporting articles. Create interactive tools such as a solar savings calculator, system sizing tool, and ROI estimator. Develop a video content strategy showcasing installations, customer testimonials, and expert interviews. Launch a monthly newsletter to engage leads and build an email subscriber base. Target 4-6 new content pieces published per month.

Technical Infrastructure (6-12 months): Implement a headless CMS for better content management and performance. Deploy automated accessibility testing in the CI/CD pipeline. Establish a Web Application Firewall with DDoS protection. Implement server-side rendering (SSR) or static site generation (SSG) for improved Core Web Vitals. Setup continuous performance monitoring with alerting thresholds.

Link Building and Digital PR (6-12 months): Develop relationships with Irish energy journalists and bloggers. Create data-driven content (Irish solar market reports, energy savings statistics) for digital PR outreach. Pursue guest posting opportunities on reputable Irish home improvement, energy, and sustainability websites. Sponsor local community events and sustainability initiatives for brand visibility and local citations.

7. Compliance and Legal Requirements

As a solar installation company operating in Ireland, Renewable Ireland must comply with a range of regulatory requirements spanning data protection, consumer protection, building regulations, and energy sector regulations. Non-compliance can result in significant financial penalties, legal action, and reputational damage.

7.1 GDPR Compliance Checklist

The General Data Protection Regulation (GDPR) requires organizations to implement comprehensive data protection measures. The following checklist evaluates the website's GDPR compliance status.

GDPR Requirement	Status	Notes
Lawful basis for data processing documented	Not Evident	Document lawful basis for each data processing activity (consent, legitimate interest, or contract).
Privacy notice / policy published	Incomplete	Privacy policy exists but may lack required information. Update to include all GDPR-mandated disclosures.
Cookie consent mechanism	Non-Compliant	Implement Category 1 CMP with granular consent. Cookie banner must not pre-tick any options.
Data subject rights facilitation	Not Evident	Provide online forms for access, rectification, erasure, portability, and objection requests.
Data breach notification process	Not Evident	Establish a breach detection and notification process. Notify DPC within 72 hours if required.
Data Processing Impact Assessment	Not Conducted	Conduct DPIA for data processing activities involving customer personal data and SEAI information.
Data retention policy defined	Not Evident	Define and document retention periods for all categories of personal data held.
Data Processing Agreements with third parties	Unknown	Ensure DPAs are in place with SolarPilot, analytics providers, CRM, and hosting providers.

Table 14. GDPR Compliance Checklist

7.2 SEAI Regulatory Requirements

The Sustainable Energy Authority of Ireland (SEAI) provides grants and administers programs for solar panel installations. Companies participating in SEAI grant schemes must meet specific requirements regarding installer qualifications, system standards, and customer information provision. The website should clearly communicate SEAI registration status, qualifications, and any relevant accreditations. SEAI requires that registered installers provide customers with accurate information about grant eligibility, application processes, and expected timelines.

Marketing materials (including website content) must not misrepresent SEAI grant amounts, eligibility criteria, or application outcomes. The website should display the SEAI registered installer logo and provide a direct link to the SEAI grant application portal. Any claims about energy savings or payback periods should be based on SEAI-published data and clearly state assumptions used in calculations.

7.3 Consumer Protection (CCPC)

The Competition and Consumer Protection Commission (CCPC) enforces consumer protection law in Ireland. Key requirements for solar installation companies include: providing clear and accurate pricing information on the website; displaying the company registration number (CRO number); clearly stating terms and conditions for installations, warranties, and cancellation policies; providing a 14-day cooling-off period for contracts concluded at a distance (online or phone); not engaging in aggressive or misleading commercial practices; and ensuring that any testimonials or reviews displayed on the website are genuine and accurately represented. The website should include a dedicated terms and conditions page, a returns/cancellation policy (where applicable), and clear pricing information for all services and installation packages. Any claims about energy savings, government grants, or product performance must be substantiated and not misleading.

7.4 Building Regulations Compliance

Solar panel installations in Ireland are subject to the Building Control Acts and related regulations. The website should provide information about building regulations requirements that customers should be aware of, including: planning permission considerations (note: most domestic solar PV installations are exempt from planning permission, but this should be clearly stated with appropriate caveats); Building Energy Rating (BER) implications of solar installations; compliance with relevant I.S. EN standards for solar thermal and photovoltaic systems; electrical safety requirements and connection to the national grid (ESB Networks); and Building Control Amendment Regulations (BCAR) where applicable. The website should recommend that customers verify compliance requirements with their local building control authority and should clearly state that Renewable Ireland installations comply with all applicable building regulations and standards.

Disclaimer: This audit report has been prepared by Z.ai Security and SEO Audit Team based on analysis conducted in April 2025. The findings, scores, and recommendations are based on the information available at the time of the audit and should be considered as a point-in-time assessment. Website conditions, security threats, search engine algorithms, and regulatory requirements may change over time. This report does not constitute legal advice. Renewable Ireland should consult with qualified legal counsel for specific compliance guidance. All recommendations should be validated in a staging environment before being applied to production systems.